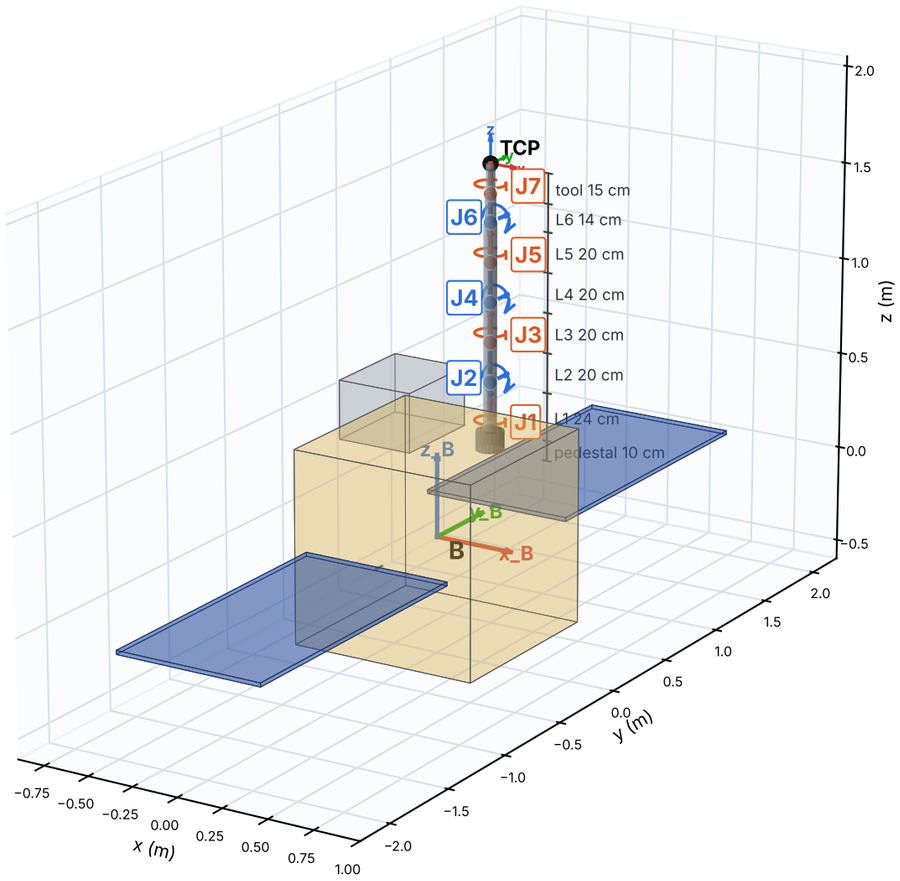
STP_Spacearm robot — 3-D kinematics diagram (7 DOF: joints J1–J7)

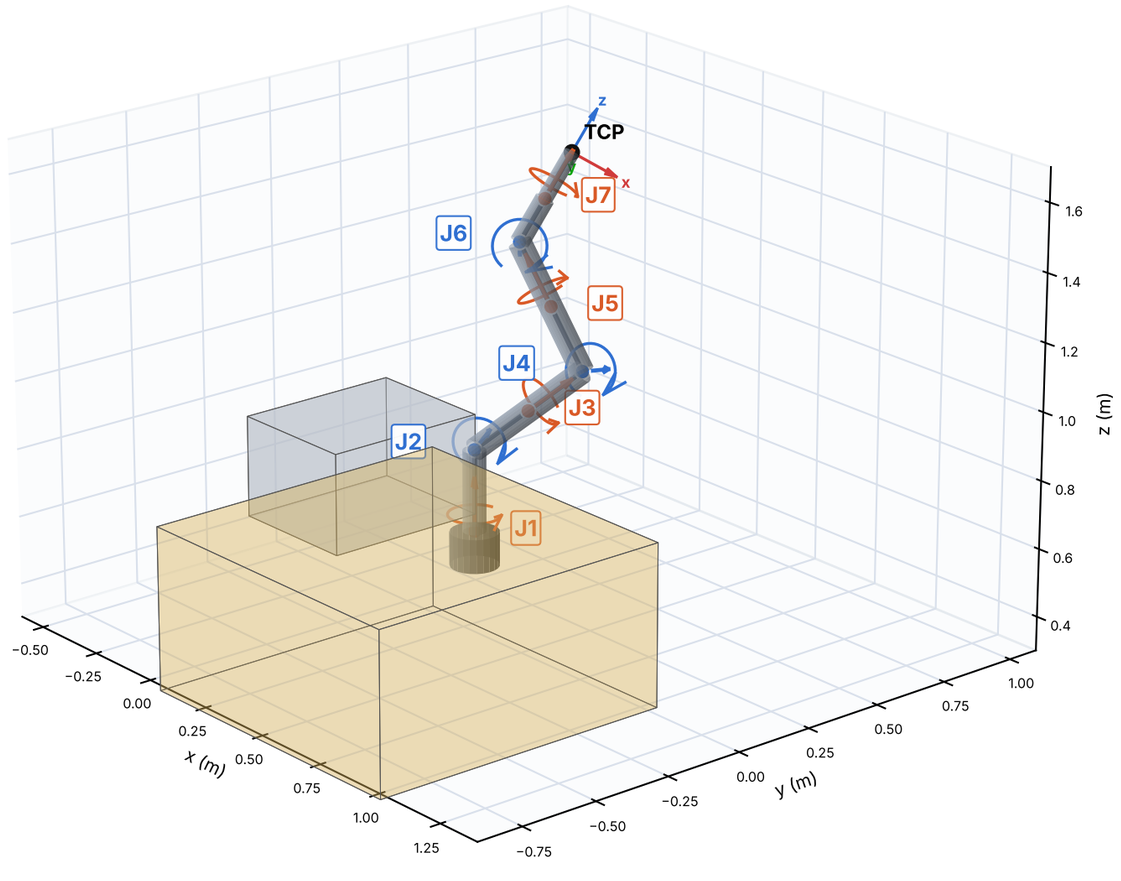
7-DOF S-R-S arm (shoulder 3 · elbow 1 · wrist 3) on a free-floating spacecraft · body frame B at the bus centre (= centre of mass), z through the arm-mount face, y along the panels

z joint: twist about the link (J1, J3, J5, J7) · y joint: bend across the link (J2, J4, J6) · x axis · y axis · z axis

Zero configuration q = 0 (arm straight up)



Example pose q = (35, 50, −20, −85, 30, 55, 0)°



TCP in frame B: (0.291, 0.358, 1.623) m (computed with the chain below)

Joint	Axis (own frame)	Motion	Offset from prev. joint	Link length	Capsule radius	Limits
J1	z	twist	10 cm	24 cm	6 cm	-170° … 170°
J2	y	bend	24 cm	20 cm	6 cm	-120° … 120°
J3	z	twist	20 cm	20 cm	6 cm	-170° … 170°
J4	y	bend	20 cm	20 cm	6 cm	-120° … 120°
J5	z	twist	20 cm	20 cm	6 cm	-170° … 170°
J6	y	bend	20 cm	14 cm	5 cm	-120° … 120°
J7	z	twist	14 cm	12 cm	4 cm	-175° … 175°

Forward kinematics (body frame B)

$t_0 = (0.30,\ 0,\ 0.50)$ m (mount), $R_0 = I$

for $i = 1..7$: $t_i = t_{i-1} + R_{i-1}\, e_z\, \ell_{i-1}$, $R_i = R_{i-1}\,\mathrm{Rot}(a_i,\ q_i)$

$\ell_0 = 10$ cm (pedestal), $\ell_1..\ell_6$ = link lengths, $a_i \in \{z, y, z, y, z, y, z\}$

$x_{TCP} = t_7 + R_7\, e_z \cdot 15$ cm · reach ≈ 1.1 m from the shoulder

7 joints but only 3 tip coordinates (position): 4 spare DOF; 7 − 6 = 1 spare even for a full pose (null space).

Source: akashrayganguly/STP_Spacearm configs/default.yaml and kinematics.py (commit a15a269).
Drawn from the configuration; none of that project's code was run.